

Writing and Solving Equations in Real-World Contexts

Teacher's Edition

Standard and Targets

MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

Boundary Forms are limited to $px \pm q = r$ and $p(x \pm q) = r$. This lesson is the first where p is routinely a fraction.

Learning targets I can write two-step equations to represent real-world contexts. I can solve two-step equations within a real-world context. I can decide whether my solution makes sense in the situation.

Essential questions How do you write a two-step equation to represent a real-world context? How do you solve a two-step equation for a real-world context, and how do you know your answer is reasonable?

Building on Lesson 6, and MA.6.AR.2.2 and MA.6.AR.2.3.

Working toward MA.8.AR.2.1 — multi-step linear equations with rational coefficients, including variables on both sides.

Where this sits in the course Fifth and last consecutive lesson on MA.7.AR.2.2, and the seventh grade 7 lesson in a row to open Unit 1. Seven days is what it takes to teach a grade 7 strand to students who have not had grade 7, and **this is the last of them — Unit 1 turns to grade 8 tomorrow**. Today adds fractional coefficients and the reasonableness check, both of which reappear immediately in the inequality lessons. Worksheet questions 9 and 10 and Additional Practice questions 9 and 10 are written at MA.8.AR.2.1; between Lessons 6 and 7 that is eight grade 8 items met a whole quarter before that benchmark is taught.

Materials Whiteboards and markers. Calculators — the worksheet divides by 0.00412 in question 8 and the contexts carry cents throughout. Worksheet 7A.01.07.

Lesson at a Glance

Windy Hill periods run **53 minutes**.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up: spaced retrieval	2	4 min
Notes I–IV: four rows every time, multiplying by $1/8$, reciprocals, which way it rounds	3–6	12 min
Example 1: a fractional coefficient	7–9	5 min
Example 2: a gift card	10–12	5 min
Example 3: two answers, both 6.4	13–15	6 min
Whiteboards: 9 questions, each with its reveal	16–33	15 min

Segment	Slides	Time
IXL — start today's skills	34	5 min
Total		53 min

Where the lesson lives: Notes II (slide 4) and Example 3 (slides 13–15). Notes II is the fraction coefficient, which is what makes this lesson different from Lesson 6. Example 3 is the reasonableness check on a problem where every line of algebra is already correct — the situation the FAST tests and the one students never see coming.

Read this before you plan the day. Math Nation's components total 45 minutes — 5 for the Warm-Up, 25 for the Collaboration, 5 for Bring It Together, 5 for Your Turn and 5 for the Wrap-Up — before any worksheet exists. The 25-minute Collaboration is five problems in tables; three of them are Examples 1 and 2 and the whiteboard sequence here, and the other two are worksheet questions 3 and 4. Bring It Together produces no mathematics at all and is replaced by Notes IV.

No partner or group work anywhere in this lesson. Math Nation's version is built on it: a 25-minute Collaboration, a gallery-walk and jigsaw suggestion, a Take Turns routine where partners alternate filling in rows of the table, a Think-Pair-Share in Bring It Together, and a Your Turn extension where early finishers write problems for each other to swap. Every one is rebuilt as individual work — the table rows are Examples 1 and 2 worked by the whole class, the Take Turns rounds are whiteboards, and the reflection is a graded exit ticket.

Vocabulary

reciprocal — the fraction turned upside down; multiplying by it undoes multiplying by the original

coefficient — the number multiplying the variable

variable — a letter standing for the quantity that changes

A note on the glossary. Math Nation lists no key terms and no additional terms for this lesson, which is a gap — the entire second half turns on undoing a fractional coefficient, and students need names for both words to talk about it. Reciprocal is carried forward from Lesson 5's board and coefficient from Lesson 3's.

Slide-by-Slide Teaching Notes

Slide 1. Writing and Solving Equations in Context *1 min*

- Post the learning target.
- This closes the equations half of the unit; inequalities start tomorrow.
- The two new ideas are small and both are places students lose points having done the algebra correctly, which is worth saying at the front.

Slide 2. Warm-Up *4 min*

- TYPE: Spaced retrieval.
- INDEPENDENT.

- ANSWERS: 1) 0.60p. 2) $x = 10$. 3) $2/5$. 4) 36. Question 4 is the prior-grade prerequisite and it is deliberately last — students who answer 4 have divided by 3, and that is precisely the error Notes II is about to name.
- OFF THE SLIDE, YOURS TO SAY: Question 4 is the one today's lesson is built on. Dividing by a fraction BELOW 1 makes things bigger.

Slide 3. Notes: I. Four rows, every time 3 min

- The four rows are Lessons 4 and 6 collected into one list, and students should be able to recite them by now.
- Write them on the board and leave them up for the rest of the lesson; every example refers back.
- OFF THE SLIDE, YOURS TO SAY: Row 4 is where the points are lost. It is not optional.

Slide 4. Notes: II. Multiplying by $1/8$ is dividing by 8 3 min

- THE NEW IDEA.
- The panel is a self-check students can actually run: $(1/4)x = 9$ must give an x bigger than 9, because a quarter of it is 9.
- Anyone landing on 2.25 has gone the wrong way, and they can catch that without being told the answer.
- OFF THE SLIDE, YOURS TO SAY: If your answer came out smaller than the total, you divided when you should have multiplied.

Slide 5. Notes: III. For any other fraction, multiply by its reciprocal 3 min

- The panel heads off the over-application of Notes II's check: $(5/2)x = 20$ gives $x = 8$, which is smaller than 20, and that is correct because $5/2$ is bigger than 1.
- The rule is about what the coefficient does, not about which direction is safe.
- OFF THE SLIDE, YOURS TO SAY: An improper coefficient makes the answer SMALLER. That is not an error.

Slide 6. Notes: IV. What the number counts decides which way it rounds 3 min

- THE OTHER NEW IDEA, and the harder of the two.
- Say the panel and then refuse to explain it until Example 3, which is the same decimal rounding both ways.
- Ordinary rounding rules are actively wrong here and students have five years of them.
- OFF THE SLIDE, YOURS TO SAY: 6.4 and 6.4 can round to 6 and to 7. The decimal does not decide.

Slide 7. Example 1 1 of the 5 min

- SPOKEN ONLY.
- Bigger — a quarter of x is 9, so x is four times that.
- Getting the prediction before the arithmetic is Notes II's check used as a tool rather than as an afterthought.

Slide 8. Example 1 2 of the 5 min

- 36 is bigger than 9, as predicted.
- Name the property — Lesson 3's table is still the vocabulary.

- Watch for students who divide by 4 and land on 2.25; the prediction they just made is what catches it.

Slide 9. Example 1 *2 of the 5 min*

- Notes III's panel, demonstrated.
- Put this beside the first half of the example: same lesson, opposite direction, and the deciding thing is whether the coefficient is bigger or smaller than 1.
- OFF THE SLIDE, YOURS TO SAY: 8 is smaller than 20, and that is right — $5/2$ is bigger than 1.

Slide 10. Example 2 *2 of the 5 min*

- Whiteboards, equation only. p = the cost of one purchase, in dollars; $60 - 5p = 12.50$.
- The rate takes money away, so its sign is negative — Lesson 6's Notes II arriving again.

Slide 11. Example 2 *2 of the 5 min*

- Both sides go negative in the second line and students who write 47.50 there get $p = -9.50$.
- The interpretation row catches it: a purchase cannot cost minus nine dollars fifty.

Slide 12. Example 2 *1 of the 5 min*

- Deliberately a slide where the answer is no.
- Students who learn 'the last step is always round' will round hours and dollars into nonsense, and asking the question and answering no is the habit, not the rounding.
- OFF THE SLIDE, YOURS TO SAY: Ask the question every time. Sometimes the answer is no, and that is still an answer.

Slide 13. Example 3 *2 of the 6 min*

- Take a vote.
- Ordinary rounding says 6 for both.
- Do not resolve it — the next slide does, and the vote is what makes it land.

Slide 14. Example 3 *2 of the 6 min*

- THE NAMED DIFFICULTY.
- Read both consequences aloud — going over budget, and leaving people behind.
- Neither is a rounding rule; both are the situation.
- This is Notes IV and it is the reason the interpretation row exists.
- OFF THE SLIDE, YOURS TO SAY: The decimal was identical. What the number counts decided everything.

Slide 15. Example 3 *2 of the 6 min*

- Lesson 4's Notes IV, returning on the last equations day.
- The instinct to round or to take the absolute value is strong and both are wrong; the answer is to re-read the setup.
- This is the sharpest two minutes in the lesson — if you only have time for one closing point, this is the one.
- OFF THE SLIDE, YOURS TO SAY: Do not round it. A negative time means the equation was set up wrong — go back and check whether the rate should have been subtracted.

Slide 16. Whiteboards · Question 1 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 17. Whiteboards · Question 1 of 9 — reveal *about 1 minute with the discussion*

- The baseline, and it is Example 1's first item on purpose: the class already has the answer, so nothing here is being diagnosed and nothing needs to be.
- It is the anchor the next five questions are varied from, and each of them changes exactly one thing about it.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 18. Whiteboards · Question 2 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 19. Whiteboards · Question 2 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the numerator is no longer 1, so dividing by the denominator alone is not enough.
- Multiply by $3/2$.
- Still bigger than the total, because $2/3$ is less than 1.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 20. Whiteboards · Question 3 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 21. Whiteboards · Question 3 of 9 — reveal *about 1 minute with the discussion*

- THE ONE THAT EARNS ITS PLACE.
- One feature changed: the fraction is improper.
- Same method, and now the answer is SMALLER than the total.
- A student holding Notes II's check as a rule rather than as a reason will reject a correct answer here.
- Deliberately NOT Example 1's improper-fraction item, which worked $(5/2)x = 20$ to 8 — this is the question the sequence exists to diagnose, and a diagnostic whose answer the class copied down ten slides ago diagnoses nothing.
- (Question 1 reuses the example on purpose; it is the anchor, not a diagnostic.) COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 22. Whiteboards · Question 4 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 23. Whiteboards · Question 4 of 9 — reveal *about 1 minute with the discussion*

- One feature changed from question 2: the total is no longer a multiple of the numerator.
- The answer stops being whole, and nothing about the method changes.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 24. Whiteboards · Question 5 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 25. Whiteboards · Question 5 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the number is now counting something.
- Round DOWN, because it is a maximum.
- Notes IV.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 26. Whiteboards · Question 6 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 27. Whiteboards · Question 6 of 9 — reveal *about 1 minute with the discussion*

- THE ONE THAT CLOSES IT.
- One feature changed from question 5: what the number counts.
- IDENTICAL decimal, opposite answer.
- Put them side by side and ask what in the arithmetic decided it — nothing did.
- The situation decided it, and no rounding rule would have got both right.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 28. Whiteboards · Question 7 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 29. Whiteboards · Question 7 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: back to solving, and the starting amount goes DOWN rather than up.
 $40 - 4c = 7.08$.
- THE BOARD TO LOOK FOR SAYS 10, from dividing 40 by 4 and never using the 7.08 — the \$7.08 is the part she did NOT spend, and leaving it out is the same shape of error as the plumber's \$75 and the giraffe's 6 feet.
- Three lessons, one habit: find the part that is left over before you divide anything.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 30. Whiteboards · Question 8 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 31. Whiteboards · Question 8 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: each child gets TWO things, so the 8 multiplies both of them — $8(15 + c) = 280$, and the bracket is doing real work.
- THE BOARD TO LOOK FOR HAS $8(15) + c = 280$ and answers \$160, which is the cash treated as one lump handed out once.
- Read the answer back into the story and it collapses: \$160 to one grandchild and nothing to the other seven.
- This is Lesson 5's bracket arriving inside a sentence instead of on a page, and the way to catch it is to ask what ONE grandchild gets before writing anything.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 32. Whiteboards · Question 9 of 9 — ask *about 90 seconds of writing*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 33. Whiteboards · Question 9 of 9 — reveal *about 2 minutes with the closing discussion*

- WRITTEN WORK, and the only question in the round that asks for it.
- With the worksheet gone this is the only place in the period the habit gets built, and IXL takes an answer without ever seeing the work.
- ANSWER: $50 \div 1.75 = 28.57\dots$, so 28 songs, costing \$49.00 and leaving \$1.00.
- THE MARK IS FOR THE REASON, not the 28.
- A student who writes "28.57 rounds to 29" has applied the rule they were taught in grade 5 and overspent the card by 75 cents — and they will be able to tell you that $29 \times 1.75 = 50.75$ if you ask, which is the point: the arithmetic was never the problem.
- This is questions 5 and 6 of this round with the answer written out.
- There is no rounding rule that gets both of those right, and there is none that gets this right either; the situation decides, and here the card decides.

- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 34. IXL 5 min

- IXL.
- The skills are assigned in Google Classroom, not named here, and they are deliberately not ours -- Croix: "The students need questions that aren't exclusively mine."
- Makes them think." Circulate.
- The thing to watch for is a student answering in the box with no paper in front of them: IXL takes an answer and never sees the work, so this is the one part of the period where the habit can quietly lapse.

Worksheet 7A.01.07 — Answer Key

1. Peyton, gift card	$40 - 4c = 7.08$, so $c = \\$8.23$ c = the cost of one purchase. The card total goes down, so the rate is negative. $4c - 40 = 7.08$ is the common wrong start.
2. Georgina, songs	$50 - 18s = 18.50$, so $s = \\$1.75$ s = the cost of one song.
3. Shirley, grandchildren	$8(15 + c) = 280$, so $c = \\$20$ c = the amount of cash each grandchild receives. Dividing both sides by 8 first is quicker than distributing. Every grandchild gets the same package, so the group repeats.
4. Fatima, running	$2m - 7 = 27$, so $m = 17$ miles Luna ran 3 miles, so Fatima ran $9(3) = 27$ this week. Work that out before writing the equation. Students who see "7 fewer" and write $7 - 2m$ have the subtraction backwards.
5. Kiara, cookies	$1\frac{2}{3}b + \frac{1}{3} = 12$, so $b = 7$ batches $(\frac{5}{3})b = 11\frac{2}{3} = \frac{35}{3}$, and multiplying by the reciprocal $\frac{3}{5}$ gives 7. Check: $1\frac{2}{3}(7) + \frac{1}{3} = 11\frac{2}{3} + \frac{1}{3} = 12$.
6. craft kit, ribbon	$\frac{1}{6}n + 2.5 = 9$, so $n = 39$ bracelets $\frac{1}{6}n = 6.5$, then multiply both sides by 6. Multiplying by $\frac{1}{6}$ is the same as dividing by 6, so the inverse is multiplying by 6 — not dividing again.
7. charter bus	$340 + 28s = 1,000$, so $s = 23.571...$ → 23 students Round DOWN. 23 students cost \$984; 24 would cost \$1,012 and break the budget. Writing 23.571 students is the error to catch.
8. water bill	$0.00412g + 22.40 = 63.60$, so $g = 10,000$ gallons This one lands on a whole number, so nothing needs rounding. Not every reasonableness check ends in a change, and students who round anyway have overcorrected.
9. two babysitters (grade 8)	$20 + 12h = 50 + 6h$, so $h = 5$ hours Subtract 6h from both sides first. Both sitters charge \$80 at 5 hours; under 5 hours Nadia is cheaper, over 5 hours Priya is.

10. $5(x - 3) = 3x + 11$
(grade 8)

$x = 13$

Distribute first: $5x - 15 = 3x + 11$, then collect. Both sides equal 50.

Question 4 is the one that needs a step before the equation. Luna ran 3 miles and Fatima ran 9 times as far, so this week's 27 has to be computed before anything is written down. Math Nation's version of this problem is a one-step equation dressed as a two-step one — $m - 7 = 9 \times 4$ collapses to $m - 7 = 36$ — so a factor of 2 was added to bring it inside the benchmark. The answer, 17, is still a whole number.

Questions 7 and 8 are a matched pair. Question 7 gives a decimal that has to be rounded down; question 8 gives an exact whole number that needs nothing. Students who round both have learned a reflex rather than a habit, and the pairing is there to catch that.

Questions 9 and 10 are grade 8 and are not expected to be fluent yet. The equation is the part to look for; the solve may still stall and that is expected. Question 10 is a bare equation on purpose — after four days of contexts, seeing $5(x - 3) = 3x + 11$ with no story around it is a useful reminder that the algebra is the same either way.

Additional Practice 7A.01.07 — Answer Key

Post this in Google Classroom alongside the practice sheet. It is optional work for students who want more.

1. Marcus, gift boxes	$5b + 6.95 = 91.70$, so $b = \\$16.95$ Shipping is charged once on the whole order, so it sits outside the group.
2. bakery, loaves	$2y - 12 = 40$, so $y = 26$ loaves The café sold 8, so the bakery sold $5(8) = 40$ today. Find the 40 before writing the equation.
3. Jamal, gift card	$75 - 6x = 13.50$, so $x = \\$10.25$
4. summer camp	$7(22 + a) = 364$, so $a = \\$30$ Each child gets the same pair of charges, so the group repeats and the parentheses belong there.
5. granola recipe	$\frac{2}{5}s + 1.4 = 7$, so $s = 14$ servings $\frac{2}{5}s = 5.6$. Multiply by the reciprocal $5/2$, or use $2/5 = 0.4$ and divide — both are fine.
6. gas bill	$0.98n + 16.50 = 114.50$, so $n = 100$ therms
7. field trip	$260 + 17s = 700$, so $s = 25.88... \rightarrow 25$ students Round DOWN. 25 students cost \$685; 26 would cost \$702.
8. Rosa's error	<i>Her first line is right. Multiplying by $\frac{1}{8}$ already divides by 8, so the inverse is multiplying by 8, not dividing again. $s = 1.25 \times 8 = 10$.</i> Math Nation names this exact error in its own misconception list for this lesson. Her own second line, $(1/8)s = 1.25$, says s is EIGHT TIMES 1.25; her 0.15625 is 1.25 divided by 8, which is 64 times too small.
9. $4(n + 5) = 6n - 2$ (grade 8)	$n = 11$ Distribute first: $4n + 20 = 6n - 2$. Moving the smaller n term keeps the coefficient positive. Both sides equal 64.
10. two tanks (grade 8)	$240 - 8t = 60 + 12t$, so $t = 9$ minutes Adding $8t$ to both sides turns the negative rate into a positive term. Both tanks hold 168 gallons at 9 minutes.

Misconceptions to Watch

- **Dividing by the denominator to undo multiplying by a unit fraction.** Trying to divide by 8 to undo multiplying by $\frac{1}{8}$. Named in Math Nation's list. Notes II, Example 1 and Additional Practice question 8. (Math Nation TE)
- **Assuming every equation has the form $px + q = r$.** Also named. Worksheet questions 3, 9 and 10 do not. (Math Nation TE)
- **Putting the subtraction backwards when a problem says "fewer."** Writing $7 - m$ for "7 fewer miles than m ." Worksheet question 4. (Math Nation TE)
- **Reporting a decimal where the situation counts whole things.** 23,571 students. Notes IV, slides 13–15, worksheet question 7.
- **Rounding an answer that was already correct.** The overcorrection after slides 13–15. Worksheet question 8 is the check.

- **Rounding the wrong direction.** Rounding a budget up or a capacity down. The whiteboard sequence.
- **Reading cents as dollars.** Using 412 instead of 0.00412 in worksheet question 8.
- **Losing the negative rate when a total goes down.** Writing $4c - 40 = 7.08$ for a gift card being spent. Worksheet questions 1 and 2.

Changes from the Math Nation Lesson

- The Component Summary was wrong and is not reproduced. It describes 1.7.1 as “Notice and Wonder about a scatter plot graph.” The actual warm-up is the Bethune-Cookman enrollment problem, and there is no scatter plot anywhere in the lesson. It also lists SE p. 42 for both 1.7.1 and 1.7.2.
- Collaboration question 4 was rewritten. As printed, $m - 7 = 9 \times 4$ collapses to $m - 7 = 36$, which is a one-step equation in a lesson whose targets are two-step equations. Doubling the relationship gives $2m - 7 = 27$ and keeps the answer a whole number.
- A cut-off answer was completed. The printed key for Collaboration question 5 reads “c = the amount of cash Shirley gives each of her” and stops there.
- The rounding instruction was dropped from the electric-bill problem. Math Nation says to round to the nearest tenth; the exact value is 980.007 kWh, the true usage is 980, and the 0.007 exists only because the \$237.86 on the bill was itself rounded to the cent. Reporting “980.0 kilowatt-hours” is less accurate than reporting 980, and their own answer line says 980.
- Two duplicates were replaced with fresh contexts. The printed Your Turn! (Kiara, $1\frac{2}{3}$ cups, save $\frac{1}{3}$) is the homework problem (Ethan, $1\frac{1}{3}$ cups, save $\frac{2}{3}$) with the numbers swapped, and the printed Additional Practice is the Collaboration with the names changed — Audrey is Shirley, and Terri is Fatima. Kiara and Shirley are kept once each on the worksheet; the Additional Practice is built on six new contexts.
- The rounding question was promoted to a full example. Example 3 is the sharpest item in the lesson — a correct equation, correct arithmetic, and an answer of 6.4 that is still wrong as an answer — and Math Nation leaves that idea in the homework with no discussion of the rounding at all.
- Notation was cleaned. The printed keys use an asterisk for multiplication ($9 * 4$, $8 * (\frac{1}{8}s)$) and contain a typo in the electric-bill equation, “0.22.358k.”
- A missing question mark was added. The printed Your Turn! ends “How many batches of cookies can Kiara make” with no punctuation, and its modeled variable definition, “b = batches of cookies,” names a unit rather than a number — in a lesson that spends a component insisting on proper definitions.
- Every partner and group activity is replaced — five of them. The 25-minute Collaboration becomes Examples 1 and 2 plus the whiteboard sequence and worksheet questions 3 and 4; the gallery-walk jigsaw is gone; Take Turns and the Think-Pair-Share are whiteboard questions; the Your Turn! swap-a-problem extension is gone.
- Bring It Together was replaced. As printed it is a reflection table keyed “Answers will vary” in both columns followed by a partner share — five minutes that produce nothing to check. Notes IV occupies that time and gives students the rounding rule the lesson’s own problems need.
- The word “homework” was removed. Math Nation’s tip reads “if time is an issue, then consider assigning Your Turn! for homework.” Worksheet 7A.01.07 is classwork; the Additional Practice goes to Google Classroom for students who want more.

- The Bethune-Cookman enrollment problem is not reproduced. Math Nation opens with it; this lesson opens with four spaced-retrieval questions instead, so nothing in the deck depends on that context or on the enrollment figure.
- Worksheet questions 9 and 10 and Additional Practice questions 9 and 10 were added at MA.8.AR.2.1 — the unknown on both sides. They are additions at the end of each sheet; no grade 7 content was cut to make room.
- Answer keys were built for both sheets from the same source file as the student versions, so the two cannot drift apart.
- Apart from Collaboration question 4 and the dropped rounding instruction, nothing in the mathematics was changed.